

JIAZHOU ZHOU

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RESEARCH INTERESTS

My research focuses on building **autonomous, adaptive and self-evolving Multimodal Agents** capable of long-horizon reasoning and continuous self-improvement in complex environments. My core interests include:

- **Multimodal Agents & Agentic RL:** Multi-turn agentic decision-making, active interrogation, on-policy distillation, and reinforcement learning for Vision-Language Models (VLMs).
- **Self-Evolving & Autonomous Systems:** Closed-loop self-reflection, test-time scaling/reasoning, and self-improving paradigms for continuous learning without dense human supervision.
- **Multimodal Reasoning & Perception Alignment:** Latent space reasoning, uncertainty estimation, and unified representation space connecting cross-modal perception.

EDUCATION

The Hong Kong University of Science and Technology (Guangzhou) Ph.D. Student in AI Thrust, Information Hub Supervisors: Lei Zhang (Chief Scientist of IDEA, IEEE Fellow), Ying-Cong Chen	09/2024–Present
University of Electronic Science and Technology of China (UESTC), China M.E. in Optical Engineering GPA: 3.82 / 4.00	06/2020–06/2023
University of Electronic Science and Technology of China (UESTC), China B.E. in Electronic Science and Technology GPA: 3.82 / 4.00	09/2016–06/2020

PROFESSIONAL & ACADEMIC EXPERIENCE

International Digital Economy Academy (IDEA) Research Intern CVR Group	08/2025–Present
The Hong Kong University of Science and Technology Research Assistant VLIS Lab Supervisor: Addison, WANG Lin	06/2023–09/2024

PUBLICATIONS (*SELECTED*)

Summary: 12 papers (10 published/accepted), featuring 4 CCF-A first-author papers (1 CVPR Highlight, 2.8%) and multiple SOTA multimodal/Agent frameworks.

Part I: Multimodal Agents, Agentic RL & On-Policy Distillation

- **J. Zhou**, H. Zhou, Y. Chen, J. Qu, Y.-C. Chen, L. Zhang. “PIVOT: Pivot-Aware On Policy Self Distillation for Multi-Turn VLM Agents.” **Under review**, 2026. | [PDF](#) | [Page](#)

Part II: Multimodal Reasoning & Latent Space Reasoning

- **J. Zhou**, Y. Chen, H. Li, Q. Jiang, H. Zhou, Y. Chen, L. Zhang. “V-Reflection: Transforming MLLMs from Passive Observers to Active Interrogators.” **Under review**, 2026. | [PDF](#) | [Page](#) | [Code](#)
- Y. Chen, Y. Yu, **J. Zhou**, Y. Shi, Y. Lan, Y. Zhang, L. Li, S. Y. Yeo. “UR²-MLLM: Uncertainty-aware Revisit Reasoning in Multimodal Large Language Models for Radiology Report Generation.” **EMNLP**, 2026. | [PDF](#)
- **J. Zhou**, X. Zheng, Y. Lyu, L. Wang. “ExACT: Language-guided Conceptual Reasoning and Uncertainty Estimation for Event-based Action Recognition and More.” **CVPR (CCF-A)**, 2024. [**Highlight (2.8%)**] | [PDF](#) | [Page](#) | [Code](#)

Part III: Multimodal Alignment, Perception & Event Vision

- L. Jiang, X. Zheng, Y. Lyu, **J. Zhou**, L. Wang. “BrightDreamer: Generic 3D Gaussian Generative Framework for Fast Text-to-3D Synthesis.” **TMLR**, 2026. | [PDF](#) | [Code](#)
- **J. Zhou**, Q. Jiang, K. Chen, L. Jiang, Y. Lyu, Y. Chen, L. Zhang. “T-Rex-Omni: Integrating Negative Visual Prompt in Generic Object Detection.” **AAAI (CCF-A)**, 2025. | [PDF](#) | [Product Page](#)
- **J. Zhou**, K. Chen, L. Zhang, L. Wang. “PASS: Path-selective State Space Model for Event-based Recognition.” **NeurIPS (CCF-A)**, 2025. | [PDF](#) | [Page](#) | [Code](#)
- Y. Lyu, X. Zheng, **J. Zhou**, L. Wang. “UniBind: LLM-Augmented Unified and Balanced Representation Space to Bind Them All.” **CVPR (CCF-A)**, 2024. | [PDF](#) | [Page](#) | [Code](#)
- K. Chen, H. Li, **J. Zhou**, Z. Wang, L. Wang. “LaSe-E2V: Towards Language-guided Semantic-Aware Event-to-Video Reconstruction.” **NeurIPS (CCF-A)**, 2024. | [PDF](#) | [Page](#)
- **J. Zhou**, X. Zheng, Y. Lyu, L. Wang. “EventBind: Learning a Unified Representation to Bind Them All for Event-based Open-world Understanding.” **ECCV (CCF-B)**, 2024. | [PDF](#) | [Page](#) | [Code](#)
- X. Zheng, Y. Lyu, **J. Zhou**, L. Wang. “Centering the Value of Every Modality: Towards Efficient and Resilient Modality-agnostic Semantic Segmentation.” **ECCV (CCF-B)**, 2024. | [PDF](#) | [Page](#)
- Z. Ping, **J. Zhou** (Co-first), W. Zhao, X. Li, L. Pu. “The edge segmentation of grains in thin-section petrographic images utilizing extinction consistency perception network.” **Complex & Intelligent Systems (SCI Q2)**, 2023. | [PDF](#) | [Code](#)

ACADEMIC SERVICES

Conference & Journal Reviewer

- Conferences: ICLR, CVPR, ECCV, ICCV, NeurIPS, AAAI, COLM, WACV, BMVC
- Journals: IEEE Transactions on Neural Networks and Learning Systems (TNNLS), IEEE Transactions on Intelligent Transportation Systems (TITS)

Teaching Assistant (HKUST-GZ)

- UFUG2102: Matrix Algebra and Applications 06/2025–07/2025
- CMAA6102: Computational Media and Arts Program Seminar 09/2024–05/2025

HONORS AND AWARDS

- NeurIPS Scholar Award 2025
- Final Winner (Top 1.5%) | ZTE Algorithm Elite Competition (Image Denoising) 2022
- CSC Outstanding Undergraduate International Exchange Program Scholarship (Top 2) 2018
- First Prize of National University Mathematical Modeling Competition (Top 10) 2018